Lead 3: n = 4, genus-4 Jacobians isogenous to Res_{F_{q⁴}/F_q} E for prime-order E

How this was produced. A Sonnet lead agent built the tool and ran every census under `PROTOCOL.md` (frozen before any run; Amendments 1, 1b, 2 and 2b are dated, and Amendment 2 froze the holdout predictions before the `q = 7` census). The parent session wrote this report from the agent's findings, because the subagent could not write report files. Every number cites a run directory under `results/`.

The parent session checked that the tool passes rustfmt, clippy `-D warnings` and its 12 unit tests. No independent validator has replayed these runs.

Theorem 6 is labelled [agent proof]. It was checked against direct linear algebra on 44 cells, but not independently reviewed.

Answer

- **Prime-order classes reached.** For prime-order `E/F_{q⁴}`, the isogeny class of `Res E` contains a smooth non-hyperelliptic genus-4 Jacobian for:
  - every admissible trace at `q = 3` (exhaustive; this reproduces round 3's 9 classes);
  - 13 of 14 traces at `q = 5` (sampled; the exception is `t = 49`);
  - all 20 traces at `q = 7` (sampled).
- **No congruence condition.** None on `(q mod ℓ, t mod ℓ)` predicts which classes are reached, and none is needed at these sizes. Tian-kernel existence does not predict the census.
- **Plane quintics.** Every curve found has degree-5 plane models over `F_q`: projection from a rational point. That is the setting of Diem's plane-curve index calculus, whose `Õ(q^{4/3})` cost (against rho's `q²`) is literature, not verified here.

What this is not. No map `Jac(C) → E` was computed, so this is existence in the isogeny class, not an attack.

Tool (`examples/genus4_n4_census.rs`)

- **Matching condition.** The Weil polynomial of `Res E` is `T⁸ − tT⁴ + q⁴`. A curve matches iff `N_k = q^k + 1` for `k = 1, 2, 3` and `N_4 = q⁴ + 1 − 4t` (Newton's identities, then Tate).
- **Curves.** Canonical models `V(Q, F) ⊂ P³` over the three quadric types (split, non-split, cone). The group orders reproduce round 3's masses 9475, 9479 and 6561 at `q = 3`, and the total `q⁹ + q⁸ − q⁶`.
- **Smoothness is exact.** `V(Q, F)` is smooth iff `J_8 = S_8`, where `J` is generated by `Q`, `F` and the 2×2 Jacobian minors.
- **Point counts.** Split quadrics and cones count roots of binary cubics. For the non-split type, even `k` is counted as for split, and odd `k` over `P¹(F_{q^{2k}})`. All three types are unit-tested against brute-force enumeration of `P³`.

Census

q	quadric type	forms searched	Res-shaped	trace allowed	smooth	mass of hits	run
3	split	43,046,721 (all)	179,184	23,040	2,304	1.000 (1 class)	`ctrlK-q3-split-run-20261024T2152Z`
3	non-split	43,046,721 (all)	274,380	31,680	2,880	1.000 (1 class)	`ctrlK-q3-nonsplit-run-20261024T2153Z`
3	cone	14,348,907 (all)	177,363	22,464	8,208	6.333 (7 classes)	`ctrlK-q3-cone-run-20261024T2151Z`
5	split	$5{\cdot}10^7$	82,699	2,476	2,476	65.6 ± 1.3	`cen-q5-split-run-20261024T2206Z`
5	non-split	$5{\cdot}10^7$	96,093	2,305	2,305	56.4 ± 1.2	`cen-q5-nonsplit-run-20261024T2210Z`
5	cone	$5{\cdot}10^7$	168,992	4,128	4,128	42.0 ± 0.7	`cen-q5-cone-run-20261024T2208Z`
7	split	$3{\cdot}10^7$	18,820	827	827	676 ± 24	`cen-q7-split-run-20261024T2239Z`
7	non-split (N4 pre-filter)	$3{\cdot}10^7$	839 (after filter)	839	839	659 ± 23	`cen-q7-nonsplit-run-20261024T2243Z`
7	cone	$3{\cdot}10^7$	34,702	949	949	217 ± 7	`cen-q7-cone-run-20261024T2241Z`

Hits per isomorphism class.

q	hits per class	classes counted against	total mass
3	$3.3{\cdot}10^{-4}$	25,515	8.33
5	$7.0{\cdot}10^{-5}$	2,328,125	164.0
7	$3.4{\cdot}10^{-5}$	46,000,759	1552

Coverage. A class with `|Aut| = 1` would be sampled about:

- at `q = 5`: 151 times (split), 41 times (non-split) and 98 times (cone), so the `q = 5` census is near-complete for classes with small `|Aut|`;
- at `q = 7`: about 1.2, 1.3 and 4.4 times, so only masses per trace and existence per trace are claimed there.

Existence per trace.

- `q = 3`: all of `{−7, −1, 9, 11}`.
- `q = 5`: 13 of 14 traces. The exception is `t = 49` (`#E = 577`, one below the Hasse bound), where the mass is $\lesssim 0.011$.
- `q = 7`: all 20 traces. The smallest is `t = 93`, with 2 hits.

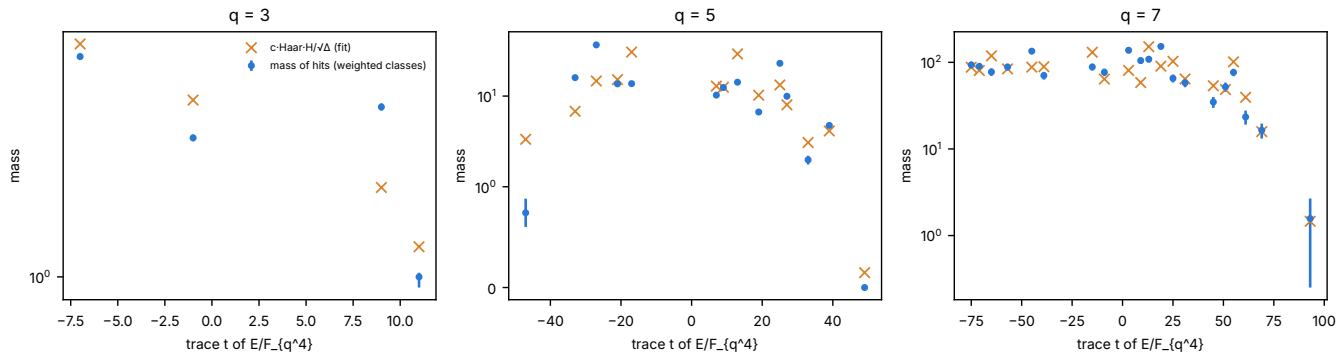


Figure 1: mass by trace

Controls

- **C2, hyperelliptic, `q = 3`, exhaustive** (`ctrlH-q3-run-20261024T2151Z`). The trace histogram is identical to round 3's, with no odd `t`.
- **C3, non-hyperelliptic, `q = 3`, exhaustive.** This reproduces round 3's 9 prime-order classes exactly: split `t = 11`, non-split `t = −1`, and cone `t = −7` (×3), `t = 9` (×3, one with `|Aut| = 3`) and `t = −1`. The mass is `8 + 1/3`.
- **C4, Theorem 5 negative control.** Hyperelliptic Res-shaped curves have no odd `t` (this control concerns hyperelliptic curves only):
  - `q = 5`: 119,496 Res-shaped among 38,397,517 squarefree forms (`ctrlH-q5-run-20261024T2156Z`);
  - `q = 7`: 55,040 among 41,985,520 (`ctrlH-q7-run-20261024T2158Z`).
- **C5.** Res-shaped curves whose trace has no elliptic curve are excluded: 66,096 of 630,927 at `q = 3`, and 24,353 of 347,784 at `q = 5`. Matching L-polynomials is necessary, not sufficient.

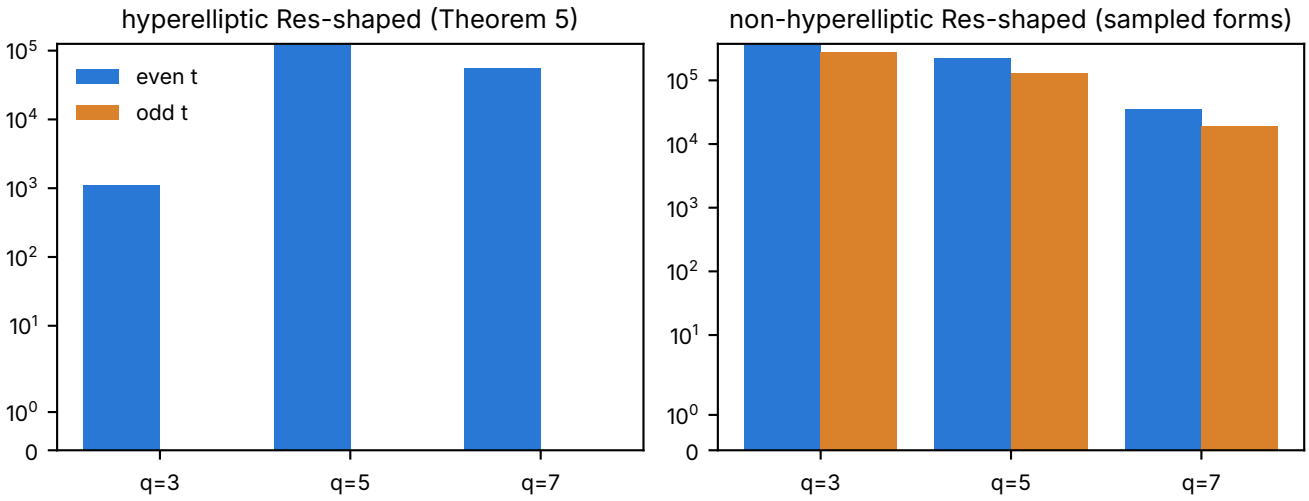


Figure 2: controls

Tian kernels and the holdout

Theorem 6 [agent proof; checked on 44 cells]. For odd ℓ `|/ q` and squarefree $\chi(T^4)$:

- factor $T^8 - tT^4 + \tilde{q}^4$ over F_ℓ;
- let $g \mapsto \tilde{g}$ send a factor to the one whose roots are $\tilde{q}/\beta$.

Then the number of Tian kernels is:

- 0 if some factor satisfies $\tilde{g} = g$;
- otherwise $2^{r/2} - 2 \cdot [\varphi \text{ split}]$, where r is the number of irreducible factors.

It agrees with direct linear algebra on all 44 regular cells at $\ell = 3, 5, 7$, with 0 disagreements (`tiantheory-run-20261024T2225Z`). Scalar and Jordan cells are tabulated only.

At $\ell = 2$ [agent claim]. For every odd `q` and every prime-order `E` there is exactly one kernel, and it is a Tian kernel. Whether the quotient is a Jacobian (the Schottky question) was not tested.

Predictors.

predictor	Spearman with mass (`q = 3, 5, 7`)	note
Tian-kernel count	−0.50 / 0.29 / 0.22	does not predict
Katz–Sarnak density `D(t)` [heuristic]	0.50 / 0.78 / 0.64	`mass = c·D`: R^2 = 0.46 at `q = 5`, 0.56 at `q = 7`
Hurwitz class number alone	0.87 / 0.57 / 0.39	poor as a fit

Holdout at `q = 7` (predictions frozen in Amendment 2):

- **P-a passed.** All 19 traces with `D ≥ 0.0143` have hits. But the agent leaned "no" on the no-claim trace `t = 93`, and it was hit. A threshold rule on `D` is not supported.
- **P-b passed**, at 0.636 and 0.217.

- **P-c failed.** The total mass is 1552, outside the predicted [490, 1100]; the forecast was a two-point `q^3.4` extrapolation.

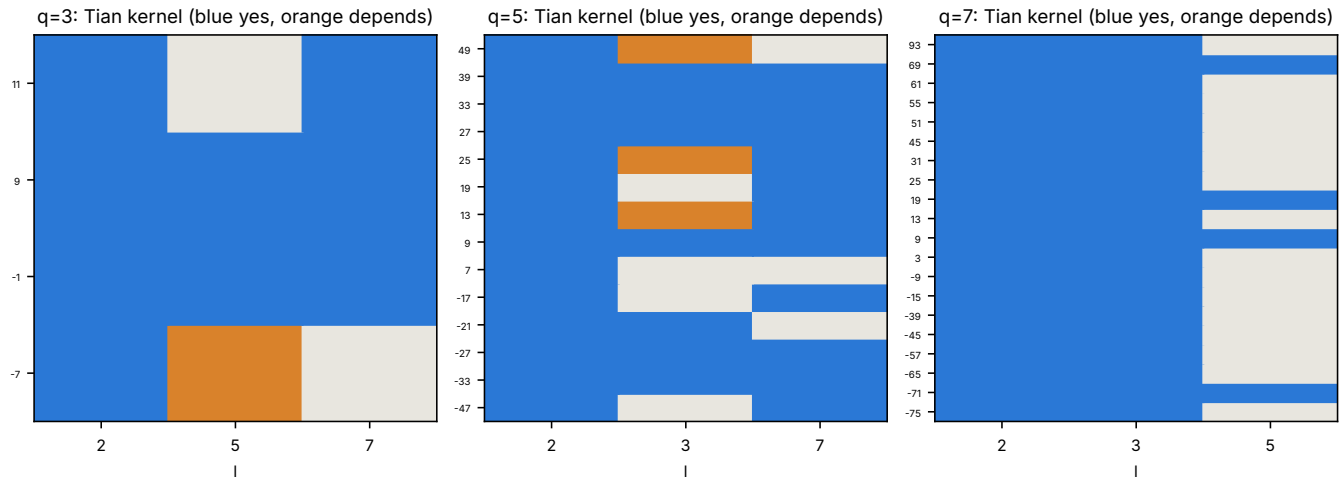


Figure 3: tian

Plane models and gonality [derived; checked]

- **No plane quartic model:** degree 4 would need a `g^1_2`.
- **Plane quintics.** The `g^2_5`'s over `F_q` are exactly `K − P` for `P ∈ C(F_q)`, so every hit, having `N_1 = q + 1 ≥ 1`, has them.
  - Projection from `P` gives the quintic `G = C1·Q2² − C2·Q2·L + C3·L²`.
  - It is singular exactly at `L = Q2 = 0`: two rational nodes for split, two conjugate nodes for non-split, one tacnode for the cone.
- **Gonality.** The `g^1_3`'s are the quadric's rulings: 2 rational (split), 1 (cone), 0 (non-split, so `F_q`-gonality 4).
- **Checks** (`plane-q{3,5,7}-run-20261024T2258Z`). 40, 408 and 928 models, with 0 point-count mismatches in 3,240 checks, and singular sets exactly as predicted.

What is not established

- **No attack.** No map `Jac(C) → E` was computed and its degree is unknown, so there is no cost statement.
- **Coverage.** Only `q = 3` is exhaustive; `q = 5` and `q = 7` are uniform samples. `q = 9` was not run, because the tool needs `q` prime.
- **The `q = 5, t = 49` zero is unexplained.** It may be a boundary effect or a rare class the sample missed.
- **The `ℓ = 2` Tian quotient** may or may not be a Jacobian; this is open.
- **Theorem 6** has not been independently reviewed.
- **Tool hashes.** The tool's sha256 differs between runs because of the dated amendments (join and theory modes); the counting code is unchanged.
- **Not yet admitted.** No independent replay has been run.

Next improvements

1. A Schottky/Igusa test of whether the `ℓ = 2` Tian quotient is a Jacobian.
2. An explicit correspondence `C → E` for one hit at `q = 3` or `q = 5`, with its degree.
3. Denser sampling near `|t| → 2q²`, and `q = 9, 11`.
4. Diem's plane-quintic index calculus on the `q = 5` hit curves, against rho in `E`.

Reproduce

`scripts/run.sh`, `scripts/plot.py`; `sh build.sh` builds this PDF.